

# Programming Assignment #2

Create your own 3D modeling tool, and explore your creativity. You will experience the entire process of geometrical modeling.

## Due:

Please check ETL

## Submission:

Please zip all files (program + supplemental material) then upload it to the ETL

## Specification:

- Basics
  - **[1pt]** Load and render patches (i.e., control points and their connections) and meshes defined in ".obj" files below. After implementing each surface, discretize the computed surfaces into triangles or quads then save them as ".obj" files, respectively. **Please zip your ".obj" files with other files (source codes, supplemental documents) when submission.**

- File:

[SNU.CG.HW2.fixed.zip](#)

- Obj description:

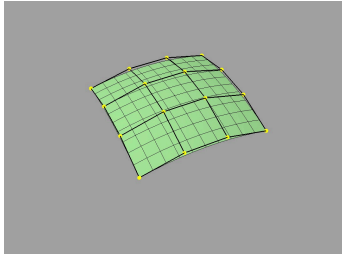
- <https://www.youtube.com/watch?v=iClme2zsg3I>
- [cs.cmu.edu/~mbz/personal/graphics/obj.html](http://cs.cmu.edu/~mbz/personal/graphics/obj.html)
- **[1pt]** Implement an interface to select and move control points (or vertices) using mouse click (for selection) and drag (for movement) directly on the screen

- Surfaces

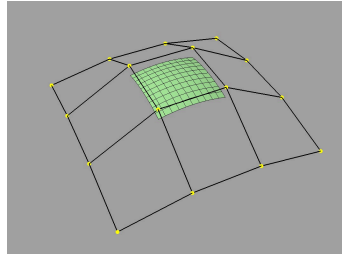
- **[2pt]** Implement bicubic Bezier surface and test on the given control points and connections
- **[2pt]** Implement uniform cubic B-Spline surface and test on the given control points and connections
- **[2pt]** Implement Catmull-Clark subdivision surface and test on the given control cage (mesh)
- **[0-2pt]** Artistic/Technical score
  - Create a demo video (<3min) that explains your program. The best demo videos will receive extra credit.
  - Guideline
    - You could emphasize your own created 3D models that are aesthetically pleasing or technically illustrative, and may show the process of creation
    - You could emphasize either extraordinary functionality or UI/UX of your 3D editing tool
  - Initially, TAs and the lecturer will select the top 10 results, respectively. From these selected results, the students will then vote for their 3 favorite submissions.
    - **[2.0pt]** 1st place
    - **[1.5pt]** 2nd-3rd place
    - **[1.0pt]** 4th-6th place
    - **[0.5pt]** 7th-10th place
- **[2pt]** Supplemental material
  - For Bezier, B-spline, Catmull-Clark subdivision surfaces:
    - ".obj" files saved from your program
    - The best screenshots of your models (this will be used for Track1)
  - For artistic/technical score evaluation, a demo video (<3min) should be included
  - The document (either Korean or English) should include:
    - Describe what you implemented, and what you haven't

- A detailed instruction on how to run and use your program
- Screenshots of your program when it runs correctly
- Any reference you used

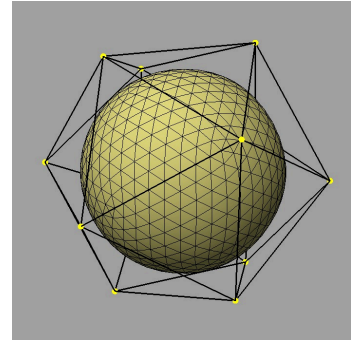
## Example Images:



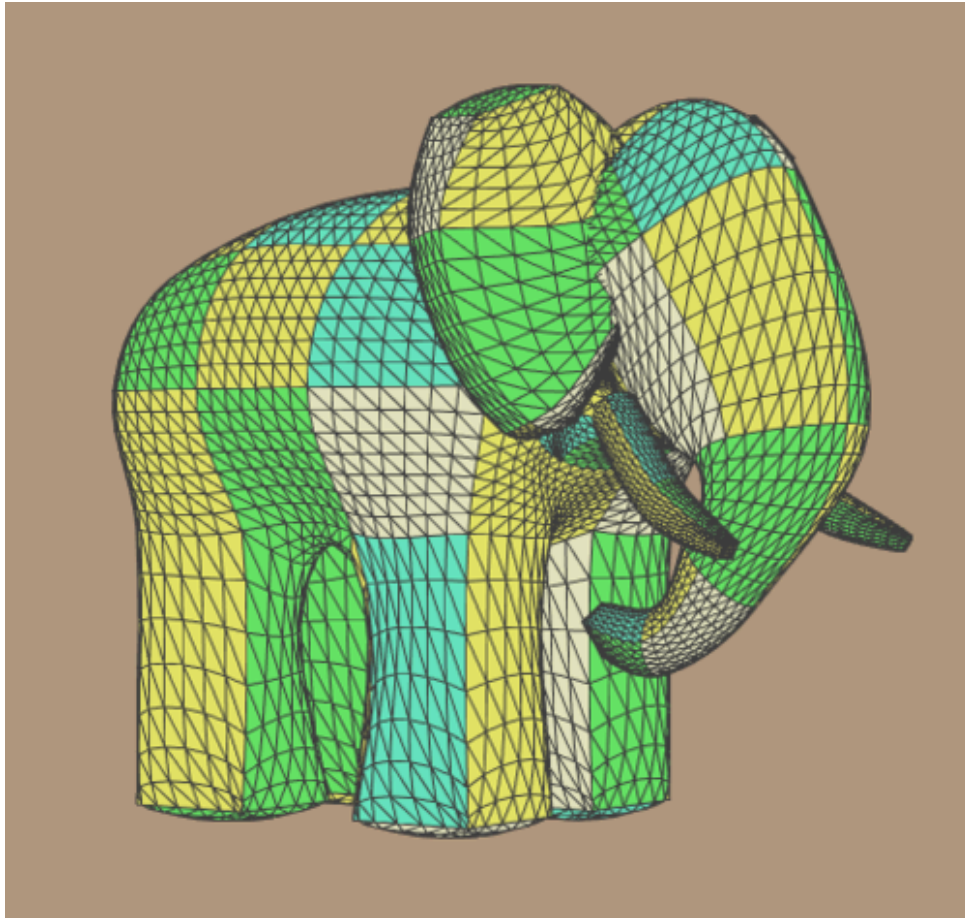
Single bicubic Bezier  
patch



Single bicubic B-Spline  
patch



Sub-division surface  
(yellow) from the original  
mesh (black lines)



Ed Catmull's "Gumbo" model, composed from Bezier patches